NEET UG STANDARD — BIOLOGY (HUMAN PHYSIOLOGY)

Chemical Coordination and Integration

Comprehensive Concept Tutorial, Endocrine Physiology, Clinical Disorders, and NEET-Aligned Practice Assessment

1. Learning Objectives and Overview

In the human body, the neural system provides a point-to-point rapid coordination among organs. However, neural coordination is fast and short-lived. Because nerve fibres do not innervate all cells of the body and cellular functions require continuous regulation, a specialized form of coordination and integration is necessary. This vital role is carried out by **hormones**. Together, the neural system and the endocrine system jointly coordinate and regulate the physiological functions of the body.

Key Learning Objectives for NEET UG

- Distinguish between the classical and current scientific definitions of hormones.
- Identify the anatomical locations and structural divisions of major human endocrine glands.
- Understand the neurosecretory relationship between the hypothalamus and the anterior and posterior pituitary.
- Master the specific hormones, target tissues, physiological roles, and clinical disorders (hypo- and hyper-secretion) of every endocrine gland.
- Recognize the endocrine functions of non-endocrine organs: the heart, kidney, and gastrointestinal tract.
- Differentiate between membrane-bound receptor action (via second messengers) and intracellular receptor action (via gene expression) based on hormone chemical nature.

2. Endocrine Glands and Hormones

Endocrine glands lack ducts and are therefore called **ductless glands**. Their secretions are called **hormones**. In contrast, exocrine glands possess ducts to release their secretions onto epithelial surfaces or into body cavities.

- **Classical Definition of a Hormone:** A chemical produced by endocrine glands, released into the blood, and transported to a distantly located target organ.
- **Current Scientific Definition of a Hormone: Hormones are non-nutrient chemicals which act as intercellular messengers and are produced in trace amounts.**

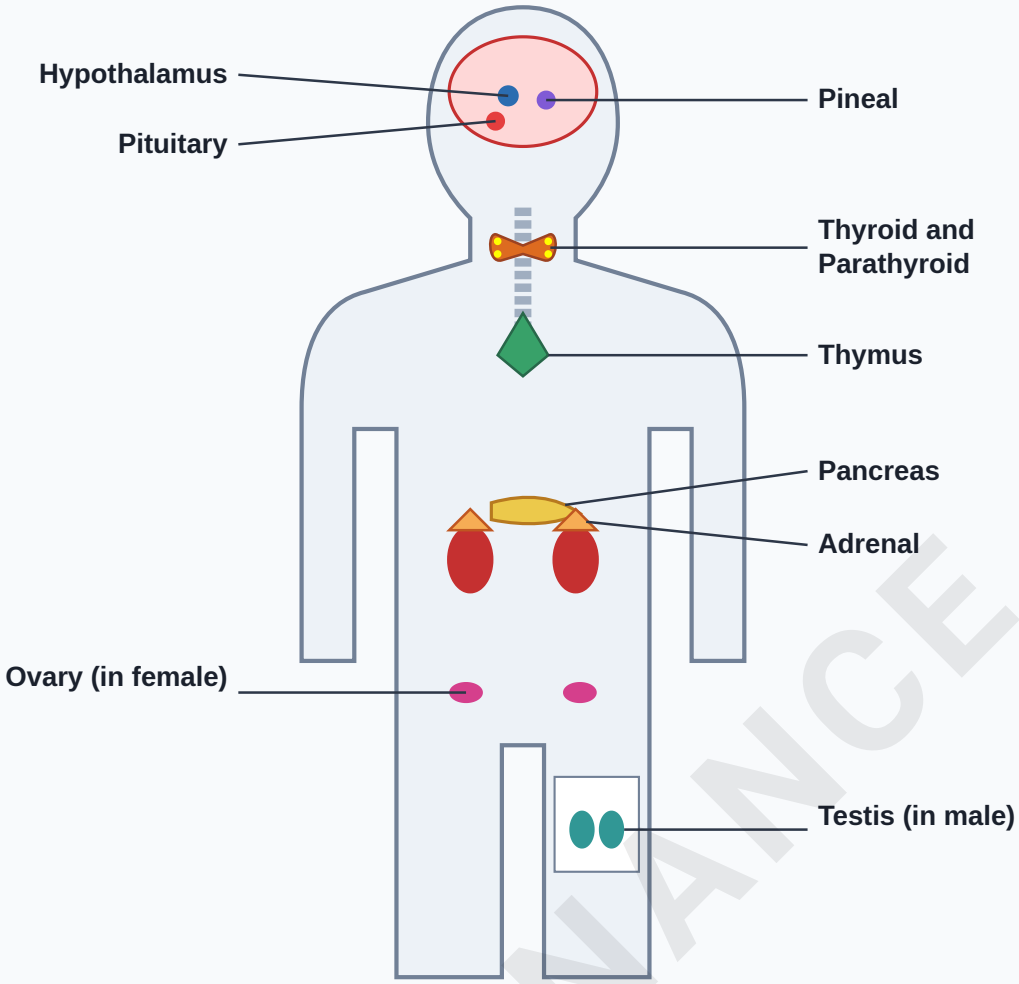
The current scientific definition is broader because it covers a number of new molecules secreted by diffused tissues and non-endocrine organs in addition to the hormones secreted by organized endocrine glands. Furthermore, invertebrates possess very simple endocrine systems with few hormones, whereas a large number of chemicals act as hormones and provide coordination in vertebrates.

3. Core Concepts: The Human Endocrine System

The endocrine system comprises organized endocrine glands as well as hormone-producing diffused tissues/cells located in different parts of the body.

- **Organized Endocrine Bodies:** Pituitary, pineal, thyroid, adrenal, pancreas, parathyroid, thymus, and gonads (testis in males and ovary in females), along with the neurosecretory hypothalamus.
- **Other Hormone-Producing Organs:** Gastrointestinal tract, liver, kidney, and heart.

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Tutorial Diagram 1: Anatomical Location of Major Endocrine Glands in the Human Body

3.1 The Hypothalamus

The **hypothalamus** is the basal part of the diencephalon (forebrain) and regulates a wide spectrum of body functions. It contains several groups of neurosecretory cells called **nuclei**, which produce hormones that regulate the synthesis and secretion of pituitary hormones.

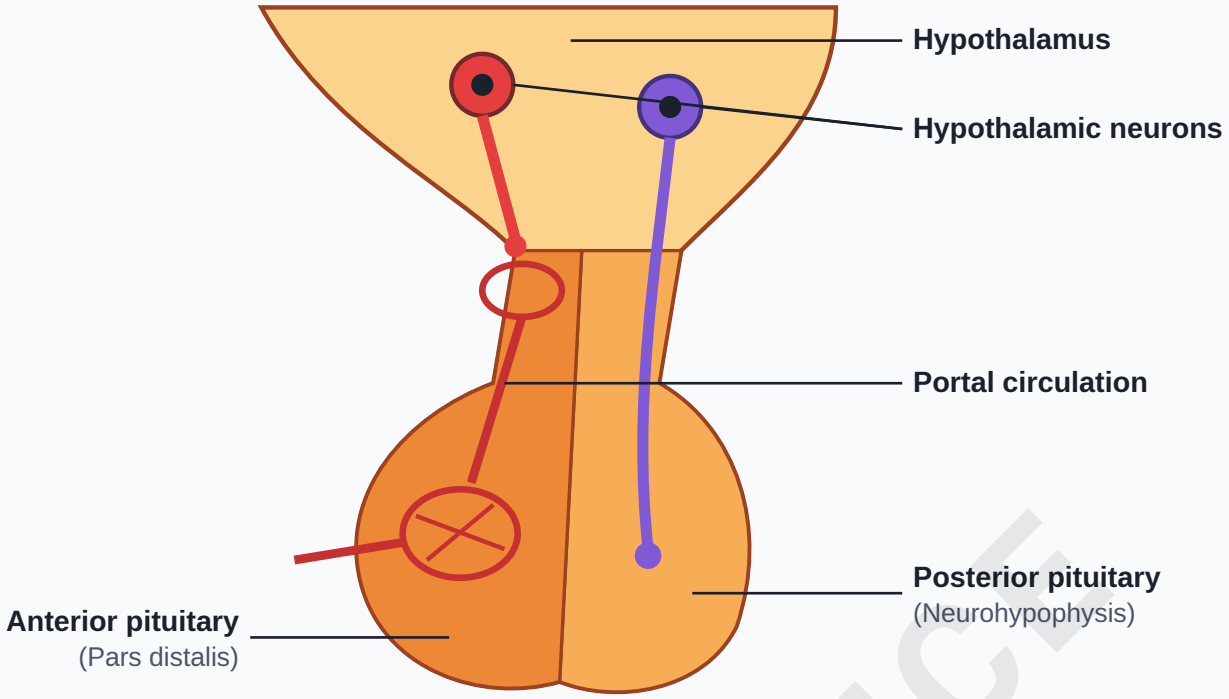
Hypothalamic hormones are of two functional types:

1. **Releasing Hormones:** Stimulate secretion of pituitary hormones. Example: **Gonadotrophin releasing hormone (GnRH)** stimulates the pituitary synthesis and release of gonadotrophins (LH and FSH).
2. **Inhibiting Hormones:** Inhibit secretions of pituitary hormones. Example: **Somatostatin** from the hypothalamus inhibits the release of growth hormone (GH) from the pituitary.

Mode of Transport to the Pituitary:

- Hormones originating in hypothalamic neurons pass through axons and are released from their nerve endings.
- **Anterior Pituitary Regulation:** Hypothalamic releasing and inhibiting hormones reach the anterior pituitary through a **hypophyseal portal circulatory system** and regulate its secretory functions.
- **Posterior Pituitary Regulation:** The posterior pituitary is under the **direct neural regulation** of the hypothalamus (hypothalamic axons extend directly into the posterior pituitary).

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Tutorial Diagram 2: Relationship Between the Hypothalamus and the Pituitary Gland

3.2 The Pituitary Gland

The pituitary gland is located in a bony cavity called **sella tursica** and is attached to the hypothalamus by a stalk. Anatomically, it is divided into two main divisions: **Adenohypophysis** and **Neurohypophysis**.

- **Adenohypophysis:** Consists of two portions:
  - **Pars distalis (Anterior Pituitary):** Produces 6 trophic/somatic hormones: Growth Hormone (GH), Prolactin (PRL), Thyroid Stimulating Hormone (TSH), Adrenocorticotropic Hormone (ACTH), Luteinizing Hormone (LH), and Follicle Stimulating Hormone (FSH).
  - **Pars intermedia:** Secretes only 1 hormone called **Melanocyte Stimulating Hormone (MSH)**. *Important NEET Note:* In humans, the pars intermedia is almost merged with the pars distalis.

- **Neurohypophysis (Pars nervosa / Posterior Pituitary):** Stores and releases 2 hormones called **oxytocin** and **vasopressin (ADH)**. These two hormones are *actually synthesised by the hypothalamus* and are transported axonally to the neurohypophysis.

Functions and Clinical Disorders of Pituitary Hormones

- **Growth Hormone (GH):** Stimulates somatic growth of the body.
  - **Gigantism:** Over-secretion of GH during childhood/growth years stimulates abnormal growth of the body.
  - **Pituitary Dwarfism:** Low secretion of GH during growth years results in stunted growth.
  - **Acromegaly:** Excess secretion of GH in adults (especially in middle age) results in severe disfigurement (especially of the face), which may lead to serious complications and premature death if unchecked. The disease is hard to diagnose in the early stages and often goes undetected for many years until changes in external features become noticeable.

- **Prolactin (PRL):** Regulates the growth of the mammary glands and formation of milk in them.
- **Thyroid Stimulating Hormone (TSH):** Stimulates the synthesis and secretion of thyroid hormones from the thyroid gland.

BOSONANCE

- **Adrenocorticotropic Hormone (ACTH):** Stimulates the synthesis and secretion of steroid hormones called **glucocorticoids** from the adrenal cortex.
- **Gonadotrophins (LH and FSH):** Stimulate gonadal activity.
  - *In Males:* **LH** stimulates the synthesis and secretion of **androgens** from the testis. **FSH** and androgens regulate **spermatogenesis**.
  - *In Females:* **LH** induces **ovulation** of fully mature follicles (Graafian follicles) and maintains the **corpus luteum** (formed from the remnants of the Graafian follicles after ovulation). **FSH** stimulates growth and development of the ovarian follicles.
- **Melanocyte Stimulating Hormone (MSH):** Acts on melanocytes (melanin-containing cells) and regulates pigmentation of the skin.
- **Oxytocin:** Acts on the smooth muscles of our body and stimulates their contraction. In females, it stimulates a vigorous contraction of the uterus at the time of childbirth, and **milk ejection** from the mammary gland.
- **Vasopressin / Anti-Diuretic Hormone (ADH):** Acts mainly at the kidney and stimulates **resorption of water and electrolytes by the distal tubules**, thereby reducing loss of water through urine (diuresis).
  - **Diabetes Insipidus:** An impairment affecting synthesis or release of ADH results in a diminished ability of the kidney to conserve water, leading to water loss (excessive dilute urine) and dehydration.

3.3 The Pineal Gland

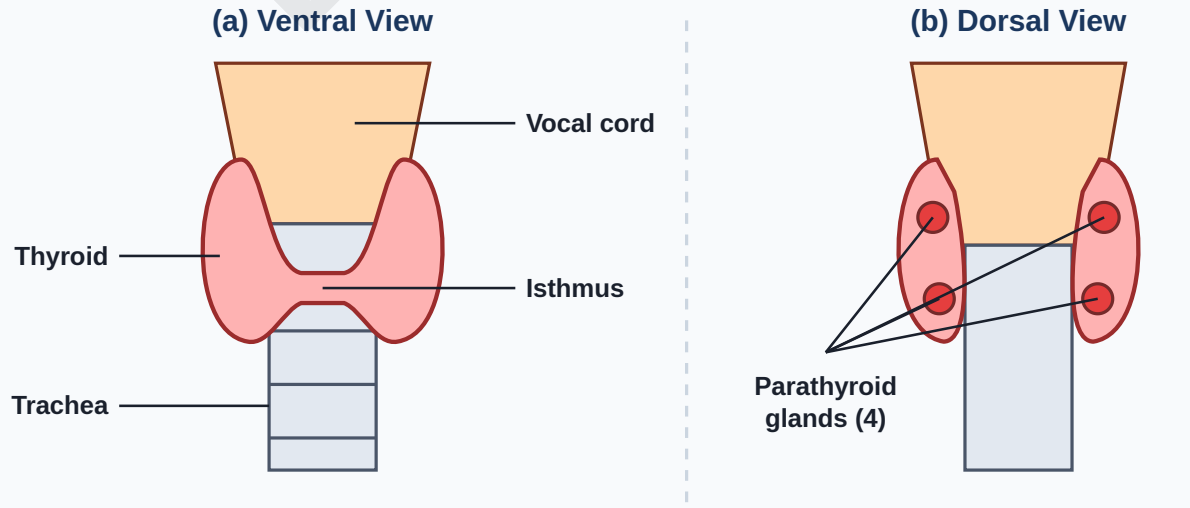
The pineal gland is located on the **dorsal side of the forebrain**. It secretes a hormone called **melatonin**.

- **Primary Role:** Regulation of a **24-hour (diurnal) rhythm** of our body — for example, maintaining the normal rhythms of the sleep-wake cycle and body temperature.
- **Additional Roles:** Melatonin also influences metabolism, pigmentation, the menstrual cycle, as well as our defense capability.

3.4 The Thyroid Gland

The thyroid gland is composed of two lobes located on either side of the trachea, interconnected by a thin flap of connective tissue called the **isthmus**. Microscopically, it consists of **follicles** and **stromal tissues**. Each thyroid follicle is composed of follicular cells enclosing a cavity.

- **Follicular Cell Hormones:** Synthesise two iodothyronine hormones: **tetraiodothyronine or thyroxine (T_4)** and **triiodothyronine (T_3)**. Iodine is essential for the normal rate of hormone synthesis in the thyroid.
- **Parafollicular / Protein Hormone:** The thyroid gland also secretes a protein hormone called **thyrocalcitonin (TCT)**, which regulates blood calcium levels by **decreasing** blood Ca^{2+} (hypocalcemic hormone).



Tutorial Diagram 3: Position of Thyroid and Parathyroid Glands — (a) Ventral Side, (b) Dorsal Side

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Functions and Clinical Disorders of the Thyroid Gland

- **Physiological Functions of T_3 and T_4:**
  1. Regulation of the **Basal Metabolic Rate (BMR)**.
  2. Support the process of **red blood cell formation (erythropoiesis)**.
  3. Control the metabolism of **carbohydrates, proteins, and fats**.
  4. Influence the maintenance of **water and electrolyte balance**.
  5. Support development and maturation of the central neural system (CNS) and normal menstrual cycle.

- **Hypothyroidism & Simple Goitre:** Deficiency of iodine in the diet results in hypothyroidism and enlargement of the thyroid gland, commonly called **goitre**.
- **Cretinism:** Hypothyroidism during pregnancy causes defective development and maturation of the growing baby, leading to stunted growth (**cretinism**), mental retardation, low intelligence quotient (IQ), abnormal skin, and deaf-mutism. In adult women, hypothyroidism may cause the menstrual cycle to become irregular.
- **Hyperthyroidism & Exophthalmic Goitre (Graves' Disease):** Cancer of the thyroid gland or development of nodules in the thyroid glands increases the rate of synthesis and secretion of thyroid hormones to abnormally high levels (hyperthyroidism). **Exophthalmic goitre (Graves' disease)** is a form of hyperthyroidism characterised by:
  - Enlargement of the thyroid gland,
  - Protrusion of the eyeballs,
  - Increased basal metabolic rate (BMR), and
  - Weight loss.

3.5 The Parathyroid Gland

In humans, **four parathyroid glands** are present on the **back (dorsal) side of the thyroid gland**, one pair each in the two lobes of the thyroid gland. They secrete a peptide hormone called **Parathyroid Hormone (PTH)**. The secretion of PTH is regulated directly by the circulating levels of calcium ions (Ca^{2+}).

- **Hypercalcemic Action of PTH:** PTH increases Ca^{2+} levels in the blood via three mechanisms:
  1. Acts on bones and stimulates the process of **bone resorption** (dissolution/demineralisation).
  2. Stimulates **reabsorption of** Ca^{2+} by the renal tubules.
  3. Increases Ca^{2+} **absorption from the digested food** (gastrointestinal tract).

- Along with TCT (thyrocalcitonin), PTH plays a significant role in calcium homeostasis in the body.

3.6 The Thymus Gland

The thymus gland is a lobular structure located **between the lungs behind the sternum on the ventral side of the aorta**. It plays a major role in the development of the immune system.

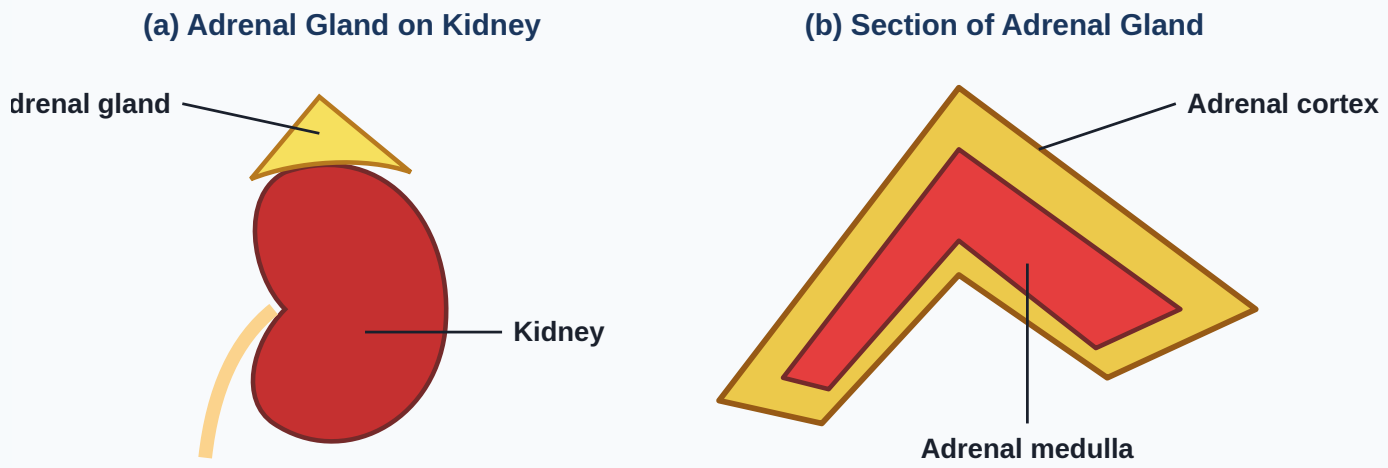
- **Hormone:** Secretes peptide hormones called **thymosins**.
- **Functions:**
  - Major role in the differentiation of **T-lymphocytes**, which provide **cell-mediated immunity (CMI)**.
  - Promotes production of **antibodies** to provide **humoral immunity**.

- **Age-Related Change:** The thymus degenerates in old individuals, resulting in a decreased production of thymosins. Consequently, the immune responses of old persons become weak.

3.7 The Adrenal Gland

Our body has one pair of adrenal glands, located **one at the anterior part (above) of each kidney**. Each gland is composed of two distinct types of tissues: the centrally located **adrenal medulla** and the outer **adrenal cortex**.

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Tutorial Diagram 4: (a) Adrenal Gland Above Kidney, (b) Section Showing Adrenal Cortex and Adrenal Medulla

1. The Adrenal Medulla

Secretes two amino-acid derivative hormones: **adrenaline (epinephrine)** and **noradrenaline (norepinephrine)**, commonly called **catecholamines**.

- **Emergency / Fight or Flight Hormones:** Rapidly secreted in response to stress of any kind and during emergency situations.
- **Physiological Effects:**
  - Increase alertness, pupillary dilation, piloerection (raising of hairs), and sweating.
  - Increase heart beat, strength of heart contraction, and rate of respiration.
  - Stimulate breakdown of glycogen (**glycogenolysis**), resulting in an increased concentration of glucose in blood.
  - Stimulate breakdown of lipids (**lipolysis**) and proteins (**proteolysis**).

2. The Adrenal Cortex

The adrenal cortex is divided into three concentric layers (from outside to inside — mnemonic **GFR**):

1. **Zona glomerulosa** (outer layer)
2. **Zona fasciculata** (middle layer)
3. **Zona reticularis** (inner layer)

The adrenal cortex secretes steroid hormones commonly called **corticoids**:

- **Glucocorticoids (Mainly Cortisol):** Involved in carbohydrate metabolism.
  - Stimulate **gluconeogenesis, lipolysis, and proteolysis**; inhibit cellular uptake and utilisation of amino acids.
  - Maintain the cardiovascular system as well as kidney functions (including blood pressure and glomerular filtration rate).
  - Produce **anti-inflammatory reactions** and **suppress the immune response**.
  - Stimulate **RBC production (erythropoiesis)**.
- **Mineralocorticoids (Mainly Aldosterone):** Regulate the balance of water and electrolytes in our body.
  - Acts mainly at the renal tubules and stimulates the **reabsorption of Na^+ and water** and the **excretion of K^+ and phosphate ions**.
  - Helps in the maintenance of electrolytes, body fluid volume, osmotic pressure, and blood pressure.

BOSONANCE

- **Androgenic Steroids:** Secreted in small amounts by the adrenal cortex; play a role in the growth of axial hair, pubic hair, and facial hair during puberty.
- **Clinical Disorder — Addison's Disease:** Underproduction of hormones by the adrenal cortex alters carbohydrate metabolism, causing acute weakness and fatigue leading to **Addison's disease**.

3.8 The Pancreas

The pancreas is a **composite (heterocrine) gland** which acts as both an exocrine and an endocrine gland. The endocrine pancreas consists of **Islets of Langerhans**. There are about 1 to 2 million Islets of Langerhans in a normal human pancreas, representing only **1 to 2 per cent** of the pancreatic tissue. The two main types of cells in the Islet of Langerhans are:

- **α-cells (Alpha cells):** Secrete the peptide hormone **glucagon**.
- **β-cells (Beta cells):** Secrete the peptide hormone **insulin**.

Comparison of Glucagon and Insulin in Glucose Homeostasis:

- **Glucagon (Hyperglycemic Hormone):** Acts mainly on liver cells (**hepatocytes**) and stimulates **glycogenolysis** and **gluconeogenesis**, resulting in increased blood sugar (**hyperglycemia**). It also reduces cellular glucose uptake and utilisation.
- **Insulin (Hypoglycemic Hormone):** Acts mainly on **hepatocytes** and **adipocytes** (cells of adipose tissue), and enhances cellular glucose uptake and utilisation. This causes a rapid movement of glucose from blood into hepatocytes and adipocytes, decreasing blood glucose levels (**hypoglycemia**). It also stimulates conversion of glucose to glycogen (**glycogenesis**) in target cells.
- **Clinical Disorder — Diabetes Mellitus:** Prolonged hyperglycemia (due to insulin deficiency and/or insulin resistance) leads to a complex disorder called **diabetes mellitus**, which is associated with **loss of glucose through urine (glycosuria)** and formation of harmful compounds known as **ketone bodies (ketonuria)**. Diabetic patients are successfully treated with insulin therapy.

3.9 The Testis

A pair of testes is present in the scrotal sac (outside the abdomen) of male individuals. The testis performs dual functions as a primary sex organ as well as an endocrine gland. It is composed of seminiferous tubules and stromal or interstitial tissue.

- **Endocrine Cells & Hormones:** The **Leydig cells (interstitial cells)**, located in the intertubular spaces, produce a group of steroid hormones called **androgens**, mainly **testosterone**.
- **Functions of Androgens:**
  1. Regulate the development, maturation, and functions of male accessory sex organs (epididymis, vas deferens, seminal vesicles, prostate gland, urethra, etc.).
  2. Stimulate muscular growth, growth of facial and axillary hair, aggressiveness, and low pitch of voice.
  3. Play a major stimulatory role in the process of **spermatogenesis** (formation of spermatozoa).
  4. Act on the central neural system (CNS) and influence male sexual behaviour (**libido**).
  5. Produce **anabolic (synthetic) effects** on protein and carbohydrate metabolism, and stimulate erythropoiesis.

3.10 The Ovary

Females have a pair of ovaries located in the abdomen. The ovary is the primary female sex organ (producing one ovum during each menstrual cycle) and also produces two groups of steroid hormones: **estrogen** and **progesterone**. The ovary is composed of ovarian follicles and stromal tissues.

- **Estrogen:** Synthesised and secreted mainly by the **growing ovarian follicles**.
  - Stimulates growth and activities of female secondary sex organs.
  - Stimulates development of growing ovarian follicles.
  - Produces appearance of female secondary sex characters (e.g., high pitch of voice, etc.) and mammary gland development.
  - Regulates female sexual behaviour.

BOSONANCE

- **Progesterone:** After ovulation, the ruptured follicle is converted into a structure called the **corpus luteum**, which secretes mainly progesterone.
  - Supports pregnancy (progestational hormone).
  - Acts on the mammary glands and stimulates the formation of **alveoli** (sac-like structures which store milk) and milk secretion.

4. Hormones of Heart, Kidney, and Gastrointestinal Tract

Several tissues and organs that are not organized endocrine glands also secrete important peptide hormones and growth factors:

Organ / Tissue	Hormone Secreted	Chemical Nature	Target & Physiological Action
Heart (Atrial wall)	**Atrial Natriuretic Factor (ANF)**	Peptide	Secreted when blood pressure increases; causes **dilation of blood vessels** (vasodilation) and thereby **reduces blood pressure**.
Kidney (Juxtaglomerular / JG cells)	**Erythropoietin**	Peptide	Stimulates **erythropoiesis** (formation of RBCs in bone marrow).
GI Tract (Endocrine cells)	**Gastrin**	Peptide	Acts on **gastric glands**; stimulates secretion of **hydrochloric acid (HCl) and pepsinogen**.
GI Tract (Endocrine cells)	**Secretin**	Peptide	Acts on the **exocrine pancreas**; stimulates secretion of **water and bicarbonate ions ($\mathrm{HCO_3^-}$)**.
GI Tract (Endocrine cells)	**Cholecystokinin (CCK)**	Peptide	Acts on both **pancreas and gall bladder**; stimulates secretion of **pancreatic enzymes and bile juice**, respectively.
GI Tract (Endocrine cells)	**Gastric Inhibitory Peptide (GIP)**	Peptide	**Inhibits gastric secretion and motility**.
Several Non-Endocrine Tissues	**Growth Factors**	Peptide / Protein	Essential for the normal growth of tissues and their **repairing/regeneration**.

5. Mechanism of Hormone Action

Hormones produce their effects on target tissues by binding to specific proteins called **hormone receptors** located in the target tissues only. Each receptor is specific to one hormone only; hence, receptors are highly specific. Binding of a hormone to its receptor leads to the formation of a **Hormone-Receptor Complex**, which triggers biochemical changes in the target tissue and regulates metabolism and physiological functions.

5.1 Classification of Hormones by Chemical Nature

1. **Peptide, Polypeptide, Protein Hormones:** Insulin, glucagon, pituitary hormones (GH, PRL, TSH, ACTH, LH, FSH, MSH, oxytocin, vasopressin), hypothalamic hormones, PTH, TCT, thymosins, ANF, erythropoietin, GI tract hormones.
2. **Steroids:** Cortisol, aldosterone, testosterone, estradiol (estrogen), progesterone.
3. **Iodothyronines:** Thyroid hormones ($\mathrm{T_3}$ and $\mathrm{T_4}$).
4. **Amino-Acid Derivatives:** Epinephrine (adrenaline), norepinephrine (noradrenaline).

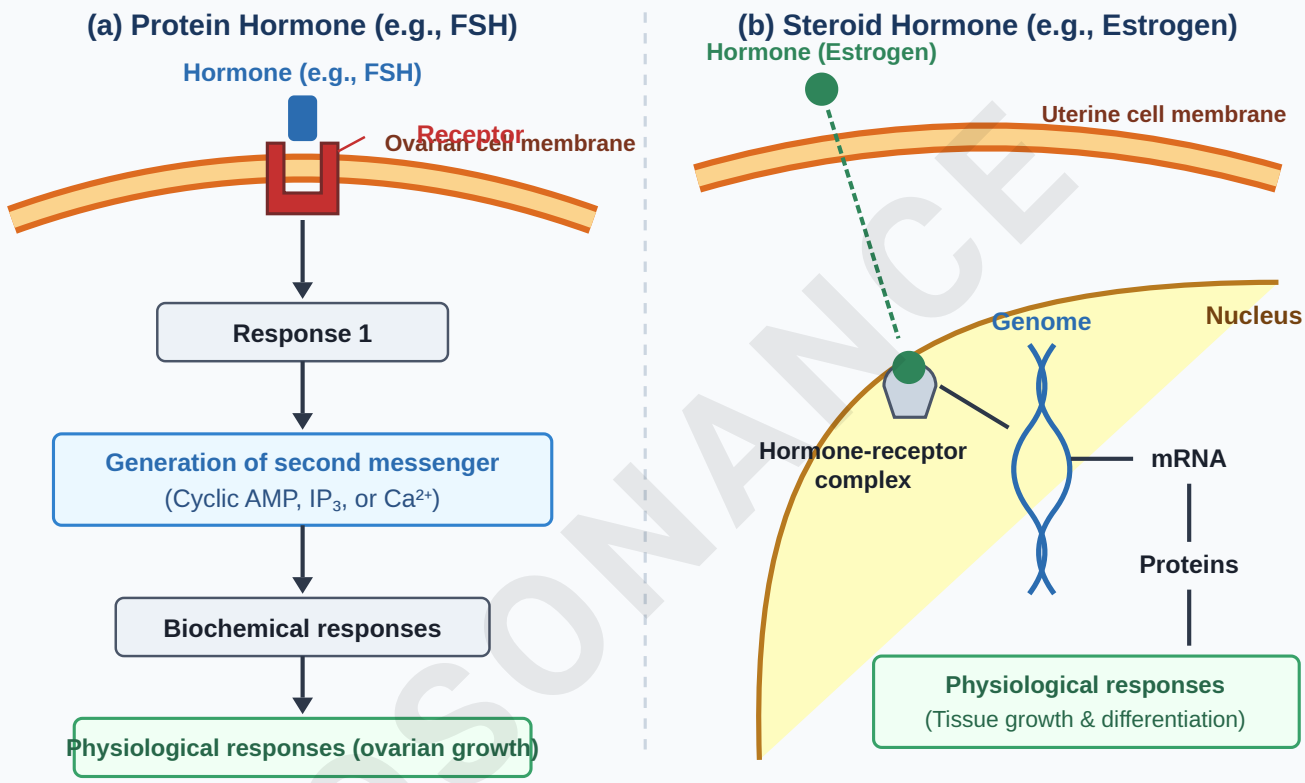
5.2 Two Modes of Receptor Interaction

- **Membrane-Bound Receptors (Extracellular / Cell Surface):**
  - *Hormone Types:* Water-soluble hormones — peptide, polypeptide, protein hormones (e.g., FSH, insulin, glucagon) and amino-acid derivatives (e.g., epinephrine).
  - *Mechanism:* These hormones **normally do not enter the target cell**. Instead, binding to the membrane receptor generates **second messengers** — such as **cyclic AMP (cAMP), $\mathrm{IP_3}$ (inositol trisphosphate),**

BOSONANCE

or Ca^{2+} — which in turn regulate cellular metabolism and cause physiological responses (e.g., ovarian growth in response to FSH).

- **Intracellular Receptors (Mostly Nuclear Receptors):**
  - *Hormone Types:* Lipid-soluble hormones — **steroid hormones** (e.g., estrogen, cortisol, testosterone, progesterone) and **iodothyronines** (T_3, T_4).
  - *Mechanism:* These hormones cross the cell membrane and enter the target cell. The hormone-receptor complex interacts directly with the **genome (DNA)** in the nucleus to **regulate gene expression or chromosome function** (transcription of $mRNA \rightarrow$ translation of proteins). Cumulative biochemical actions result in physiological and developmental effects (e.g., tissue growth and differentiation).



Tutorial Diagram 5: Mechanism of Hormone Action — (a) Protein Hormone via Membrane Receptor & Second Messenger, (b) Steroid Hormone via Intracellular Nuclear Receptor & Gene Expression

6. Master Reference Tables & Antagonistic Hormone Pairs

6.1 Antagonistic (Opposing) Hormone Pairs & Synergistic Regulators

Physiological Parameter	Hormone Increasing Parameter	Hormone Decreasing Parameter
Blood Glucose Level	**Glucagon** (Pancreatic α-cells), **Cortisol**, **Epinephrine**	**Insulin** (Pancreatic β-cells) — *Hypoglycemic*
Blood Ca^{2+} Level	**Parathyroid Hormone (PTH)** — *Hypercalcemic*	**Thyrocalcitonin (TCT)** — *Hypocalcemic*
Blood Pressure / Vessel Tone	**Aldosterone**, **Vasopressin (ADH)**, **Catecholamines**	**Atrial Natriuretic Factor (ANF)** — *Vasodilator*
Anterior Pituitary GH Release	Growth Hormone Releasing Hormone (Hypothalamus)	**Somatostatin** (Hypothalamus)
Gastric Secretion & Motility	**Gastrin** (GI Tract)	**Gastric Inhibitory Peptide (GIP)** (GI Tract)

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High-Yield NEET Connection: Multiple Hormones Supporting Erythropoiesis (RBC Formation)

Four different hormones/groups explicitly mentioned in the syllabus support or stimulate red blood cell formation (**erythropoiesis**):

1. **Erythropoietin** (from Juxtaglomerular cells of the kidney)
2. **Thyroid hormones (T_3, T_4)** (from the thyroid gland)
3. **Cortisol** (glucocorticoid from the adrenal cortex)
4. **Androgens (Testosterone)** (from Leydig cells of the testis)

7. Worked Examples (Chapter Concept Applications)

Worked Example 1: Target Glands of Trophic Hormones

Problem: Identify the specific target gland or target cells for each of the following hormones:

a. Hypothalamic hormones
b. Thyrotrophin (TSH)
c. Corticotrophin (ACTH)
d. Gonadotrophins (LH, FSH)
e. Melanotrophin (MSH)

Solution:

- **(a) Hypothalamic hormones:** Pituitary gland (specifically anterior pituitary via portal circulation).
- **(b) Thyrotrophin (TSH):** Thyroid gland.
- **(c) Corticotrophin (ACTH):** Adrenal cortex.
- **(d) Gonadotrophins (LH, FSH):** Gonads — Testis (in males) and Ovary (in females).
- **(e) Melanotrophin (MSH):** Melanocytes (melanin-containing cells in the skin).

BOSONANCE

Worked Example 2: Functional Classification of Hormones

Problem: Give specific example(s) of each of the following functional categories of hormones:

a. Hyperglycemic hormone and hypoglycemic hormone
b. Hypercalcemic hormone
c. Gonadotrophic hormones
d. Progestational hormone
e. Blood pressure lowering hormone
f. Androgens and estrogens

Solution:

- **(a) Hyperglycemic hormone:** Glucagon; **Hypoglycemic hormone:** Insulin.
- **(b) Hypercalcemic hormone:** Parathyroid hormone (PTH).
- **(c) Gonadotrophic hormones:** Luteinizing hormone (LH) and Follicle stimulating hormone (FSH).
- **(d) Progestational hormone:** Progesterone.
- **(e) Blood pressure lowering hormone:** Atrial Natriuretic Factor (ANF).
- **(f) Androgens:** Testosterone; **Estrogens:** Estradiol.

BOSONANCE

Worked Example 3: Differential Diagnosis of Endocrine Deficiencies

Problem: Which specific hormonal deficiency is responsible for each of the following clinical conditions, and how does Diabetes Mellitus differ from Diabetes Insipidus?

a. Diabetes mellitus
b. Simple Goitre
c. Cretinism
d. Diabetes insipidus
e. Addison's disease

Solution:

- **(a) Diabetes mellitus:** Deficiency of **insulin** (secreted by β-cells of the Islets of Langerhans in the pancreas) and/or insulin resistance.
- **(b) Simple Goitre:** Deficiency of **iodine** in the diet leading to deficiency of thyroid hormones (T_3 and T_4 / **hypothyroidism**).
- **(c) Cretinism:** Deficiency of **thyroid hormones (hypothyroidism) during pregnancy**, causing stunted growth and mental retardation in the growing baby.
- **(d) Diabetes insipidus:** Impairment affecting synthesis or release of **Anti-Diuretic Hormone (ADH / Vasopressin)** from the hypothalamus/neurohypophysis.
- **(e) Addison's disease:** Underproduction of **corticoids (hormones of the adrenal cortex)**.

Key Clinical Distinction: Both *Diabetes mellitus* and *Diabetes insipidus* involve excessive urination (diuresis), but **Diabetes mellitus** is caused by insulin deficiency and features **glucose and ketone bodies in the urine** with hyperglycemia, whereas **Diabetes insipidus** is caused by ADH deficiency and features watery, glucose-free urine with dehydration.

8. NEET UG Practice Questions with Worked Solutions

Note: The following original practice questions are strictly aligned with the NEET UG examination standard to test conceptual clarity, multi-statement analysis, assertion-reasoning, and clinical correlation.

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Practice Question 1 (Single-Choice MCQ — Concept Definition)

According to the current scientific definition, hormones are best described as:

(1) Nutrient chemicals produced exclusively by ductless glands in large quantities.

(2) Non-nutrient chemicals which act as intercellular messengers and are produced in trace amounts.

(3) Biocatalysts secreted into ducts to regulate intracellular enzyme kinetics.

(4) Nutrient lipophilic molecules produced only by organized neurosecretory nuclei.

Correct Answer: (2)

Worked Solution: While the classical definition restricted hormones to chemicals produced by organized endocrine glands and transported via blood to distant target organs, the current scientific definition states that **hormones are non-nutrient chemicals which act as intercellular messengers and are produced in trace amounts**. This definition covers molecules secreted by diffused tissues and non-endocrine organs as well.

Practice Question 2 (Single-Choice MCQ — Neuroendocrine Regulation)

Which of the following statements accurately describes the anatomical and functional relationship between the hypothalamus and the pituitary gland?

(1) Both the anterior pituitary and posterior pituitary receive hypothalamic hormones through a portal circulatory system.

(2) Oxytocin and vasopressin are synthesised in the pars nervosa and transported to the hypothalamus via portal veins.

(3) Hypothalamic hormones reach the anterior pituitary through a portal circulatory system, whereas the posterior pituitary is under the direct neural regulation of the hypothalamus.

(4) Somatostatin is a hypothalamic hormone that stimulates the release of growth hormone from the neurohypophysis.

Correct Answer: (3)

Worked Solution: Releasing and inhibiting hormones originating in hypothalamic neurons pass through axons, are released at nerve endings, and reach the anterior pituitary (pars distalis) through a **portal circulatory system**. In contrast, the **posterior pituitary (neurohypophysis) is under the direct neural regulation of the hypothalamus**, storing and releasing oxytocin and vasopressin which are actually synthesised in the hypothalamus and transported axonally.

BOSONANCE

Practice Question 3 (Multi-Statement Evaluation — Pituitary Disorders)

Read the following statements regarding pituitary hormones and their clinical disorders:

- **Statement I:** Excess secretion of growth hormone in adults, especially in middle age, causes severe disfigurement (especially of the face) called Acromegaly.
- **Statement II:** In humans, the pars intermedia is completely separate from the pars distalis and secretes prolactin and MSH.
- **Statement III:** An impairment affecting synthesis or release of ADH results in Diabetes Mellitus, characterised by loss of glucose through urine.
- **Statement IV:** In males, LH stimulates the synthesis and secretion of androgens from the testis.

Which of the above statements are **correct**?

(1) Statements I and IV only

(2) Statements I, II, and IV only

(3) Statements II and III only

(4) Statements I, III, and IV only

Correct Answer: (1)

Worked Solution:

- **Statement I is correct:** Excess GH in adults (middle age) causes Acromegaly (severe facial disfigurement, premature death if unchecked).
- **Statement II is incorrect:** In humans, the pars intermedia is almost merged with the pars distalis, and it secretes only one hormone (MSH); prolactin is secreted by the pars distalis.
- **Statement III is incorrect:** Impairment of ADH synthesis or release causes **Diabetes Insipidus** (water loss and dehydration, without glycosuria), not Diabetes Mellitus.
- **Statement IV is correct:** LH acts on Leydig cells in the male testis to stimulate androgen synthesis and secretion.

BOSONANCE

Practice Question 4 (Single-Choice MCQ — Thyroid & Parathyroid Physiology)

A patient presents with enlargement of the thyroid gland, protrusion of the eyeballs, increased basal metabolic rate, and weight loss. This clinical condition is identified as:

(1) Cretinism, a form of hypothyroidism during pregnancy

(2) Exophthalmic goitre (Graves' disease), a form of hyperthyroidism

(3) Addison's disease, caused by underproduction of adrenal cortical hormones

(4) Acromegaly, caused by hypersecretion of TSH in middle age

Correct Answer: (2)

Worked Solution: Exophthalmic goitre (also called **Graves' disease**) is a form of **hyperthyroidism** characterised by enlargement of the thyroid gland, protrusion of the eyeballs, increased basal metabolic rate (BMR), and weight loss.

Practice Question 5 (Assertion-Reason — Calcium Homeostasis)

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Parathyroid hormone (PTH) is classified as a hypercalcemic hormone.

Reason (R): PTH stimulates bone resorption (demineralisation), increases Ca^{2+} reabsorption by renal tubules, and increases Ca^{2+} absorption from digested food.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

Correct Answer: (1)

Worked Solution: PTH increases blood Ca^{2+} levels and is therefore a hypercalcemic hormone (Assertion A is true). It achieves this increase through three specific mechanisms: stimulating bone resorption/demineralisation, stimulating Ca^{2+} reabsorption by the renal tubules, and increasing Ca^{2+} absorption from digested food in the gut (Reason R is true and directly explains Assertion A).

BOSONANCE

Practice Question 6 (Single-Choice MCQ — Thymus & Immunity)

Why do the immune responses of old individuals become weak compared to young individuals?

(1) Over-secretion of cortisol in old age completely destroys the bone marrow.

(2) The thymus gland degenerates in old individuals, resulting in a decreased production of thymosins.

(3) The pineal gland calcifies and stops producing melatonin, which is the sole hormone responsible for T-lymphocyte differentiation.

(4) Parathyroid hormone levels drop sharply in old age, halting antibody synthesis.

Correct Answer: (2)

Worked Solution: The thymus gland secretes peptide hormones called **thymosins**, which play a major role in the differentiation of T-lymphocytes (cell-mediated immunity) and also promote antibody production (humoral immunity). Because the **thymus degenerates in old individuals**, thymosin production decreases, weakening their immune responses.

Practice Question 7 (Single-Choice MCQ — Adrenal Cortex Layers & Hormones)

Which of the following correctly lists the three layers of the adrenal cortex from the **innermost layer to the outermost layer**, along with a correct function of cortisol?

(1) Zona glomerulosa $\rightarrow$ Zona fasciculata $\rightarrow$ Zona reticularis; Cortisol inhibits RBC production.

(2) Zona reticularis $\rightarrow$ Zona fasciculata $\rightarrow$ Zona glomerulosa; Cortisol produces anti-inflammatory reactions and suppresses the immune response.

(3) Zona fasciculata $\rightarrow$ Zona reticularis $\rightarrow$ Zona glomerulosa; Cortisol stimulates cellular uptake and utilisation of amino acids.

(4) Zona reticularis $\rightarrow$ Zona glomerulosa $\rightarrow$ Zona fasciculata; Cortisol acts mainly at renal tubules to excrete Na^{+} ions.

Correct Answer: (2)

Worked Solution: Pay close attention to the direction requested: from **innermost to outermost**, the adrenal cortex layers are **zona reticularis (inner layer)** $\rightarrow$ **zona fasciculata (middle layer)** $\rightarrow$ **zona glomerulosa (outer layer)**. Furthermore, cortisol (the main glucocorticoid) produces anti-inflammatory reactions, suppresses the immune response, stimulates gluconeogenesis, lipolysis, proteolysis, and RBC production, and *inhibits* cellular uptake and utilisation of amino acids.

BOSONANCE

Practice Question 8 (Single-Choice MCQ — Pancreatic Hormones)

How do insulin and glucagon jointly maintain glucose homeostasis in the blood?

(1) Insulin stimulates glycogenolysis in hepatocytes, whereas glucagon stimulates glycogenesis in adipocytes.

(2) Insulin enhances cellular glucose uptake and glycogenesis (causing hypoglycemia), whereas glucagon stimulates glycogenolysis and gluconeogenesis in hepatocytes (causing hyperglycemia).

(3) Both insulin and glucagon are steroid hormones that act via nuclear receptors in pancreatic acinar cells.

(4) Glucagon is secreted by β-cells to lower blood sugar, while insulin is secreted by α-cells to raise blood sugar.

Correct Answer: (2)

Worked Solution: Insulin (secreted by β-cells) acts on hepatocytes and adipocytes to enhance cellular glucose uptake and utilisation and stimulates conversion of glucose to glycogen (**glycogenesis**), lowering blood glucose (hypoglycemia). Glucagon (secreted by α-cells) acts on hepatocytes to stimulate **glycogenolysis** and **gluconeogenesis** while reducing cellular glucose uptake, raising blood glucose (hyperglycemia).

BOSONANCE

Practice Question 9 (Matching Type — GI Tract, Heart, and Kidney Hormones)

Match the hormones listed in **Column I** with their specific source and physiological action in **Column II**:

Column I (Hormone)	Column II (Source & Action)
(A) Cholecystokinin (CCK)	(i) Atrial wall of heart; causes dilation of blood vessels to reduce blood pressure
(B) Secretin	(ii) Juxtaglomerular cells of kidney; stimulates erythropoiesis
(C) Atrial Natriuretic Factor (ANF)	(iii) GI tract; acts on exocrine pancreas to stimulate secretion of water and bicarbonate ions
(D) Erythropoietin	(iv) GI tract; acts on both pancreas and gall bladder to stimulate secretion of pancreatic enzymes and bile juice

Choose the correct option:

(1) A-(iv), B-(iii), C-(i), D-(ii)

(2) A-(iii), B-(iv), C-(ii), D-(i)

(3) A-(iv), B-(i), C-(iii), D-(ii)

(4) A-(ii), B-(iii), C-(i), D-(iv)

Correct Answer: (1)

Worked Solution:

- **(A) CCK:** Acts on both pancreas (pancreatic enzymes) and gall bladder (bile juice) $\rightarrow$ (iv).
- **(B) Secretin:** Acts on exocrine pancreas to stimulate secretion of water and bicarbonate ions (HCO_3^-) $\rightarrow$ (iii).
- **(C) ANF:** Secreted by atrial wall of the heart; dilates blood vessels to lower blood pressure $\rightarrow$ (i).
- **(D) Erythropoietin:** Produced by juxtaglomerular cells of the kidney; stimulates erythropoiesis $\rightarrow$ (ii).

BOSONANCE

Practice Question 10 (Single-Choice MCQ — Chemical Nature & Mechanism of Hormone Action)

Which of the following pairs of hormones **do NOT** enter their target cells, but instead bind to membrane-bound receptors and generate second messengers such as cAMP, $\mathrm{IP_3}$, or $\mathrm{Ca^{2+}}$?

(1) Estrogen and Progesterone

(2) Thyroxine ($\mathrm{T_4}$) and Cortisol

(3) Follicle Stimulating Hormone (FSH) and Insulin

(4) Testosterone and Triiodothyronine ($\mathrm{T_3}$)

Correct Answer: (3)

Worked Solution: Peptide/protein hormones (such as **FSH, insulin, glucagon, pituitary hormones**) and amino-acid derivatives (epinephrine) interact with **membrane-bound receptors**, normally do not enter the target cell, and generate second messengers ($\mathrm{cAMP}, \mathrm{IP_3}, \mathrm{Ca^{2+}}$). In contrast, steroid hormones (estrogen, progesterone, cortisol, testosterone) and iodothyronines ($\mathrm{T_3}, \mathrm{T_4}$) cross the cell membrane and bind to **intracellular (mostly nuclear) receptors** to regulate gene expression.

Practice Question 11 (Statement-Based MCQ — Gonadal Hormones)

Consider the following two statements regarding gonadal hormones in humans:

Statement A: Estrogen is synthesised and secreted mainly by the growing ovarian follicles, whereas progesterone is secreted mainly by the corpus luteum formed after ovulation.

Statement B: Progesterone acts on the mammary glands to stimulate the formation of alveoli (sac-like structures which store milk), whereas oxytocin stimulates milk ejection from the mammary glands.

Choose the correct option:

(1) Both Statement A and Statement B are correct.

(2) Both Statement A and Statement B are incorrect.

(3) Statement A is correct but Statement B is incorrect.

(4) Statement A is incorrect but Statement B is correct.

Correct Answer: (1)

Worked Solution: Both statements are completely accurate. Growing ovarian follicles mainly synthesise and secrete estrogen; after ovulation, the ruptured Graafian follicle forms the corpus luteum (maintained by LH), which secretes progesterone. On the mammary glands, prolactin regulates growth and milk formation, progesterone stimulates alveoli formation and milk secretion, and oxytocin stimulates milk ejection.

BOSONANCE

9. Common Exam Pitfalls & Revision Checklist

High-Frequency NEET Pitfalls to Avoid

- **Pitfall 1 — Synthesis vs. Storage of Posterior Pituitary Hormones:** Never state that the posterior pituitary (neurohypophysis) *synthesises* oxytocin and vasopressin. They are **synthesised by the hypothalamus** and only stored and released by the neurohypophysis.
- **Pitfall 2 — Milk Formation vs. Alveoli Formation vs. Milk Ejection:**
  - Growth of mammary glands & formation of milk $\rightarrow$ **Prolactin (PRL)**.
  - Formation of alveoli (sac-like milk storage structures) & milk secretion $\rightarrow$ **Progesterone**.
  - Milk ejection from mammary glands $\rightarrow$ **Oxytocin**.
- **Pitfall 3 — Thyroid Hormones ($\mathrm{T_3}/\mathrm{T_4}$) Receptor Location:** Even though $\mathrm{T_3}$ and $\mathrm{T_4}$ are iodothyronines (derived from an amino acid), they are lipid-soluble and act via **intracellular (nuclear) receptors** to regulate gene expression, *unlike* epinephrine which acts via membrane-bound receptors.
- **Pitfall 4 — Order of Adrenal Cortex Layers:** Read carefully whether the exam question asks from *outer to inner* (Zona glomerulosa $\rightarrow$ Zona fasciculata $\rightarrow$ Zona reticularis) or from *inner to outer* (Zona reticularis $\rightarrow$ Zona fasciculata $\rightarrow$ Zona glomerulosa).
- **Pitfall 5 — Secretin vs. CCK Targets: Secretin** acts on the exocrine pancreas to release *water and bicarbonate ions*; **CCK** acts on *both* the pancreas (pancreatic enzymes) and the gall bladder (bile juice).

Final NEET Revision Checklist

- ☑ Can you state the current scientific definition of a hormone word-for-word (non-nutrient chemicals, intercellular messengers, trace amounts)?
- ☑ Do you know the exact anatomical locations of the pituitary (sella tursica), pineal (dorsal forebrain), thyroid/parathyroid, and thymus (ventral side of aorta behind sternum)?
- ☑ Can you distinguish between Gigantism, Pituitary Dwarfism, and Acromegaly (GH disorders)?
- ☑ Can you contrast Cretinism, Simple Goitre (hypothyroidism), and Exophthalmic Goitre / Graves' disease (hyperthyroidism)?
- ☑ Do you remember all four hormones that stimulate erythropoiesis (Erythropoietin, Thyroid hormones, Cortisol, Androgens)?
- ☑ Can you trace the steps of membrane-bound receptor action (FSH $\rightarrow$ cAMP/$\mathrm{IP_3}$/$\mathrm{Ca^{2+}}$ $\rightarrow$ ovarian growth) versus nuclear receptor action (Estrogen $\rightarrow$ genome/mRNA/proteins $\rightarrow$ uterine growth)?

10. Concise Final Recap

Chemical coordination and integration in humans is achieved through an intricate network of organized endocrine glands (hypothalamus, pituitary, pineal, thyroid, parathyroid, thymus, adrenal, pancreas, testis, ovary) and hormone-secreting tissues in the heart (ANF), kidney (erythropoietin), and gastrointestinal tract (gastrin, secretin, CCK, GIP). The hypothalamus links the nervous and endocrine systems by regulating the anterior pituitary via a portal system and the posterior pituitary via direct axons. Water-soluble peptide and catecholamine hormones act via cell-membrane receptors and second messengers ($\mathrm{cAMP}, \mathrm{IP_3}, \mathrm{Ca^{2+}}$), whereas lipid-soluble steroid and iodothyronine hormones act via intracellular nuclear receptors to regulate gene expression—maintaining metabolic homeostasis, growth, stress adaptation, immune competence, and reproduction.

BOSONANCE